

Module Theory

Warm Up

Exercise 1.

- (a) The element 2 is invertible in both $\mathbb{Z}[\frac{1}{2}]$ and in $\mathbb{Z}/3\mathbb{Z}$. Find a homomorphism $\phi: \mathbb{Z}[\frac{1}{2}] \rightarrow \mathbb{Z}/3\mathbb{Z}$ which sends $2^{-1} \mapsto 2^{-1}$.
- (b) Find an element in $\ker \phi$ which is not in $\mathbb{Z}$.

Exercise 2. Let $R = \mathbb{Z}/12\mathbb{Z}$. Compute the localization R_f for each $f \in \{2, 3, 4, 5, 6\}$.

Exercise 3. (August 2021 Problem 1 parts (a) and (b))

- (a) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization $R_f = 0$?
- (b) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization R_f a field? (Recall that the zero ring is not a field.)

Exercise 4. (January 2020 Problem 3) Let R be a commutative ring, I an ideal in R , and S a multiplicative subset of R . There are two standard correspondences involving prime ideals:

- (i) Prime ideals in R/I are in bijection with prime ideals in R which contain I .
 - (ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S .
- (a) Prove (i) and (ii). (The original problem said “or”, but in the SEP you have no choice.)
 - (b) Give an example of a ring with finitely many elements that has exactly three prime ideals.
 - (c) Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

Exercise 5.

- (a) Prove that *localization is exact*. That is, if $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is an exact sequence of R -modules, then $0 \rightarrow S^{-1}A \rightarrow S^{-1}B \rightarrow S^{-1}C \rightarrow 0$ is an exact sequence of $S^{-1}R$ -modules.
- (b) If B is an R -module, and A is a submodule, prove that $S^{-1}(B/A) \cong S^{-1}B/S^{-1}A$. That is, you can localize or quotient in any order.
- (c) Compute the isomorphism types of the following:

$$\left(\frac{\mathbb{C}[x, y]}{(xy)} \right)_x \quad \text{and} \quad \left(\frac{\mathbb{C}[x, y]}{(xy)} \right)_{x+y}.$$

Exercise 6.

- (a) Prove that a $\mathbb{C}[x]$ -module is the same thing as a $\mathbb{C}$ -vector space V and a linear transformation $X: V \rightarrow V$.
- (b) Suppose $V = \mathbb{C}^n$, and $X: V \rightarrow V$ is a linear transformation. Then (V, X) is a finitely-generated $\mathbb{C}[x]$ -module, so it is isomorphic to $\bigoplus_i \mathbb{C}[x]/(f_i(x))$ for some polynomials $f_i(x)$. In linear algebra terms, what is the relation between the polynomials f_i and the linear transformation X ?

Work Out

Exercise 7. (January 2022 Problem 3) Let R be a commutative ring.

- (a) Prove that a module M over R is projective if and only if it is a direct summand of a free module.
- (b) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$ -module.

Exercise 8. (January 2021 Problem 3) Two questions about projective modules.

- (a) Show that a projective module over an integral domain is torsion free.
- (b) Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:
 - (i) If M_1 and M_2 are projective, then so is M_3 .
 - (ii) If M_2 and M_3 are projective, then so is M_1 .

Exercise 9. (January 2025 Problem 5, parts (a), (c), (d)) Let A be a commutative Noetherian ring, and M an A -module. We say that M has finite length if there exists a positive integer n such that every chain

$$M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_k$$

satisfies $k \leq n$. The smallest such n is then called the length of M .

- (a) Prove that if M has finite length, then so does every localization $M_{\mathfrak{p}}$ at a prime ideal $\mathfrak{p}$ of A .
- (c) Prove that the following converse of (a) holds if A has finitely many maximal ideals: under this assumption, M has finite length if $M_{\mathfrak{m}}$ has finite length for every maximal ideal $\mathfrak{m}$ of A . (You can use without proof the fact that an A -module M is zero if and only if $M_{\mathfrak{m}}$ is zero for every maximal ideal $\mathfrak{m}$ of A .)
- (d) Prove that the assumption in part (c) on the finiteness of the number of maximal ideals is necessary by providing an example of a ring A and an A -module M which is not of finite length, even though $M_{\mathfrak{m}}$ is of finite length for every maximal ideal $\mathfrak{m}$ of A . (Hint: you can find such an example where A is the ring of integers.)